10 AWS Incident Nightmares Every SRE Should Know

Real Production Failure Patterns and How to Investigate Them

This quick guide highlights common failure patterns that regularly occur in AWS-based production systems. Recognising these patterns early can dramatically reduce incident resolution time.

For deeper investigation techniques and copy-paste CloudWatch queries, see the AWS Log Search Recipes handbook.

1. API Gateway 502 / 504 Storms

Symptoms: Sudden spikes of 502 or 504 responses.
Typical Cause: Downstream service timeout or integration failure.
First Check: Backend service latency and integration logs.

2. Lambda Cold Start Failures

Symptoms: First requests after deployment fail or are slow.
Typical Cause: Dependency loading or initialization delays.
First Check: Lambda init duration and startup logs.

3. DynamoDB Throttling Cascades

Symptoms: Latency increases followed by retry storms.
Typical Cause: Exceeded provisioned capacity.
First Check: DynamoDB throttle metrics and retry logs.

4. Retry Amplification Loops

Symptoms: Traffic spike during partial outages.
Typical Cause: Aggressive retry logic without backoff.
First Check: Repeated retry log entries for the same request.

5. Partial Deployments

Symptoms: Some instances behave differently from others.
Typical Cause: Incomplete deployment rollout.
First Check: Instance version identifiers in logs.

6. Dependency API Outages

Symptoms: Internal services returning generic errors.
Typical Cause: Third-party or internal API failure.
First Check: Upstream timeout or connection errors.

7. Configuration Drift

Symptoms: Environment behaves differently despite no new deploy.
Typical Cause: Manual configuration changes.
First Check: Configuration values logged at startup.

8. Silent Data Corruption

Symptoms: Incorrect results but no errors.
Typical Cause: Schema mismatch or bad transformations.
First Check: Validation or mismatch warnings in logs.

9. Resource Exhaustion

Symptoms: Containers restarting or instances disappearing.
Typical Cause: Memory or CPU exhaustion.
First Check: OOM and resource pressure messages.

10. Queue Backlogs and Processing Lag

Symptoms: System appears healthy but outputs are delayed.
Typical Cause: Consumer slowdown or stuck workers.
First Check: Queue lag and worker processing logs.

Investigate Faster with AWS Log Search Recipes

The AWS Log Search Recipes handbook contains practical CloudWatch Logs Insights queries designed for real production incidents.

Get the full toolkit here:

Incident Engineering Patterns